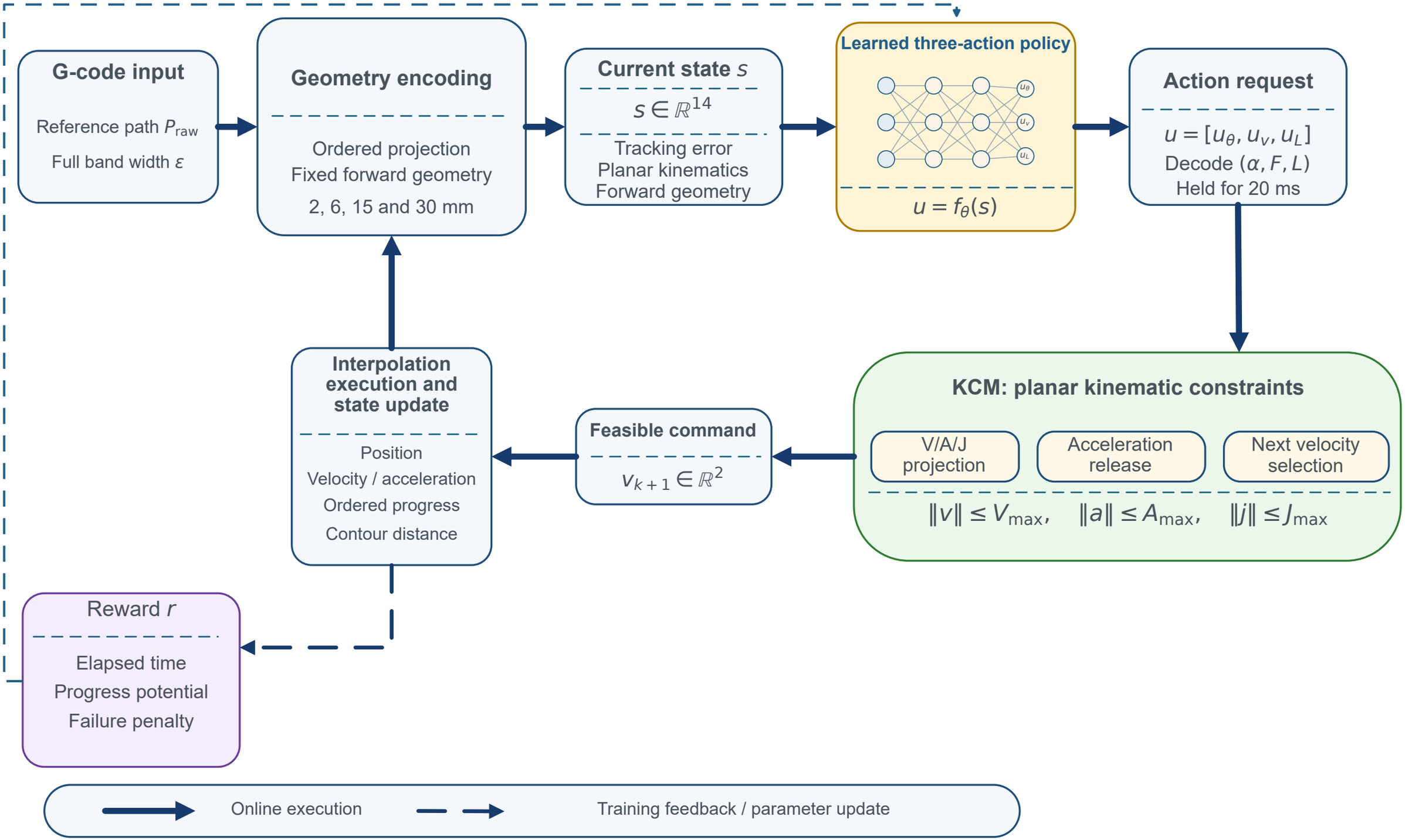

G-code input
Reference path P_{raw}
Full band width ε
Geometry encoding
Ordered projection
Fixed forward geometry
2, 6, 15 and 30 mm
Current state s
$s \in \mathbb{R}^{14}$
Tracking error
Planar kinematics
Forward geometry
Learned three-action policy
u_θ
u_v
u_L
$u = f_\theta(s)$
Action request
$u = [u_\theta, u_v, u_L]$
Decode (α, F, L)
Held for 20 ms
KCM: planar kinematic constraints
V/A/J projection
Acceleration release
Next velocity selection
$\|v\| \le V_{max}, \quad \|a\| \le A_{max}, \quad \|j\| \le J_{max}$
Feasible command
$v_{k+1} \in \mathbb{R}^2$
Interpolation execution and state update
Position
Velocity / acceleration
Ordered progress
Contour distance
Reward r
Elapsed time
Progress potential
Failure penalty
Online execution
Training feedback / parameter update